

NOAA Northeast Fisheries Science Center: Gulf of Maine - Aerial Surveys 2024-2027

Type Enterprise Geodatabase Feature Class

Tags NOAA, NEFSC, Aerial Surveys, Marine Mammals, Whales, GOM

Summary

This layer depicts transects from the NOAA-Northeast Fishery Science Center’s aerial surveys that are conducted from 2024-2027 in the Gulf of Maine area. This layer contains six separate track lines that make up NEFSC’s survey effort for the entire GOM area. These lines show only the planned routes for the aerial surveys and are displayed for planning and research coordination purposes. Realized routes can be found on <https://whalemap.org/>.

Description

This layer depicts planned transects from NOAA-Northeast Fishery Science Center’s aerial surveys conducted from 2024-2027 in the Gulf of Maine. For more information on methodology and project details visit: <https://www.fisheries.noaa.gov/new-england-mid-atlantic/marine-mammal-protection/whale-and-dolphin-research-northeast#north-atlantic-right-whale-sighting-surveys> ; and the RWSC database project page: <https://database.rwsc.org/details?recordId=rec5rsRHi5cVbXmAk>

Credits

Tim Cole - NOAA [tim.cole@noaa.gov](https://mailto:tim.cole@noaa.gov)

Use limitations

These data will be used by RWSC and its expert Subcommittees, partners, and other participants to implement the [Science Plan](#), including to understand the extent of ongoing and planned data collection activities, and to coordinate and plan future data collection and research activities with respect to offshore wind. The data will be displayed via online mapping platforms.

Extent

West -70.624715 East -67.041869
North 44.585919 South 40.985000

Scale Range

Maximum (zoomed in) 1:5,000
Minimum (zoomed out) 1:150,000,000

Topics and Keywords ▶

Themes or categories of the resource Oceans

Content type ⇔ Downloadable Data
Export to FGDC CSDGM XML format as Resource Description No

Citation ▶

Title NOAA Northeast Fisheries Science Center: Gulf of Maine - Aerial Surveys 2024-2027
Creation date 2025-05-02 00:00:00
Publication date 2025-08-19 00:00:00

Presentation formats ⇔ digital map

Citation Contacts ▶

Responsible party - point of contact
Individual's name Tim Cole
Organization's name NOAA

Contact information ▶
Phone
Voice NA
Address
Type postal
Delivery point NA
City NA
Administrative area NA
Postal code NA
e-mail address tim.cole@noaa.gov

Responsible party - originator
Individual's name Deborah Brill
Organization's name RWSC
Contact's position Marine Mammals Subcommittee Lead

Contact information ▶
Address
Type
e-mail address deborah.brill@duke.edu

Resource Details ▶

Dataset languages ⇔ English (UNITED STATES)
Dataset character set utf8 - 8 bit UCS Transfer Format

Status on-going

Spatial representation type ⇔ vector

Supplemental information

This layer depicts transects from NOAA-Northeast Fishery Science Center's aerial surveys conducted from 2024-2027 in the Gulf of Maine. The NEFSC's survey of the GOM area is split into six separate effort areas with individual track lines, although sometimes multiple areas are completed within a single outing. The areas include the Coast of Maine, Franklin Basic, Great South Channel, Jordan Basin, Rodgers Basin and Wilkinson Basin. Data collection focuses on six large whale species - blue, fin, sei, humpback, sperm, and North Atlantic right whales - but also includes opportunistic sightings of sea turtles and other marine animals. The objectives are: (1) to determine each whale species' distribution and estimate density; (2) to determine each whale species' monthly and seasonal occurrence; (3) to characterize the interannual variability of the whales' distributions and densities; and (4) to record behavioral data of sighted whales as much as possible. Summaries will be publicly available as surveys progress.

These surveys are a continuation of the NEFSC Right Whale Aerial Survey program that has been running since 1998. The program began with patrols of Great South Channel and transitioned in 2002 to systematic line surveys of the northeastern United States Exclusive Economic Zone from the Gulf of Maine to South of Martha's Vineyard and Nantucket. In 2006, the program transitioned to box surveys in the Gulf of Maine and Southern New England in order to increase the number of right whales identified per flight hour. In 2018, box survey efforts began south of New England. These box surveys continued through 2023, with special attention to right whale aggregations within the boxes. In 2024 the surveys were extended to the current coverage and converted to the line surveys seen in the layer.

The survey team used a de Havilland Twin Otter aircraft flown at 750 to 1500 feet at 100-110 knots. The surveys along the Coast of Maine are conducted twice a year, (once in January/February and once in May/June), the Great South Channel route is flown four times a year (once in each quarter) and the Wilkinson, Jordan, Rogers, and Franklin Basin routes are conducted six times a year (approx. every two months).

These lines show only the planned routes for the aerial surveys and are displayed for planning and research coordination purposes. Realized routes can be found on <https://whalemap.org/>.

For more information visit the project site:<https://www.fisheries.noaa.gov/new-england-mid-atlantic/marine-mammal-protection/whale-and-dolphin-research-northeast#north-atlantic-right-whale-sighting-surveys>; and the RWSC database project page: <https://database.rwsc.org/details?recordId=rec5rsRH5cVbXmAk>

Processing environment ⇔ Microsoft Windows 10 Version 10.0 (Build 26200) ; Esri ArcGIS 13.7.0.1901

Credits

Tim Cole - NOAA <[a href=https://mailto:tim.cole@noaa.gov target="_blank">tim.cole@noaa.gov](mailto:tim.cole@noaa.gov)

ArcGIS item properties

Name ⇔ rpt.rpt.NEFSC_GOM_Aerial_Surveys
Size ⇔ 0.000
Location ⇔ Server=rwsc-db-pg15.env.duke.edu; Service=sde:postgresql:rwsc-db-pg15.env.duke.edu; Database=rpt; User=rpt; Version=sde.DEFAULT
Access protocol ⇔ ArcSDE Connection

Extents ►

Extent

Description
Aerial Surveys conducted from 2024 - 2027

Geographic extent

Bounding rectangle
Extent type
Extent used for searching
West longitude -70.624715
East longitude -67.041869
North latitude 44.585919
South latitude 40.985000

Temporal extent

Beginning date 2024-01-01 00:00:00
Ending date 2027-12-31 00:00:00

Vertical extent

Minimum value ⇔ 0.000000
Maximum value ⇔ 0.000000

Extent in the item's coordinate system

westBL ⇔ -70.624715
eastBL ⇔ -67.041869
southBL ⇔ 40.985000
northBL ⇔ 44.585919
exTypeCode ⇔ Yes

Resource Points of Contact ►

Point of contact - point of contact

Individual's name Tim Cole
Organization's name NOAA

Contact information ►

Phone
Voice NA
Address
Type postal
Delivery point NA
City NA
Administrative area NA
Postal code NA
e-mail address tim.cole@noaa.gov

Resource Maintenance ►

Resource maintenance
Update frequency as needed

Resource Constraints ▶

Constraints
Limitations of use
These data will be used by RWSC and its expert Subcommittees, partners, and other participants to implement the [Science Plan](#), including to understand the extent of ongoing and planned data collection activities, and to coordinate and plan future data collection and research activities with respect to offshore wind. The data will be displayed via online mapping platforms.

Spatial Reference ▶

ArcGIS coordinate system
Type ⇔ Geographic
Geographic coordinate reference ⇔ GCS_WGS_1984
Coordinate reference details ⇔
GeographicCoordinateSystem
WKID 4326
XOrigin -400
YOrigin -400
XYScale 1111948722.2222219
ZOrigin -100000
ZScale 10000
MOrigin -100000
MScale 10000
XYTolerance 8.983152841195215e-09
ZTolerance 0.001
MTolerance 0.001
HighPrecision true
LeftLongitude -180
LatestWKID 4326
WKT
GEOGCS["GCS_WGS_1984",DATUM["D_WGS_1984",SPHEROID["WGS_1984",6378137.0,298.257223563]],PRIMEM["Greenwich",0.0],UNIT["Degree",0.0174532925199433],AUTHORIT

Reference system identifier
Value ⇔ 4326
Codespace ⇔ EPSG
Version ⇔ 6.2(3.0.1)

Spatial Data Properties ▶

Vector ▶
Level of topology for this dataset ⇔ geometry only

Geometric objects
Feature class name rpt.rpt.NEFSC_GOM_Aerial_Surveys
Object type ⇔ composite
Object count ⇔ 6

ArcGIS Feature Class Properties ▶
Feature class name rpt.rpt.NEFSC_GOM_Aerial_Surveys
Feature type ⇔ Simple
Geometry type ⇔ Polyline
Has topology ⇔ FALSE
Feature count ⇔ 6
Spatial index ⇔ TRUE
Linear referencing ⇔ TRUE

Data Quality ▶

Data quality report - Conceptual consistency ▶
Data quality measure reference
Measure description
Data represents planned aerial survey routes with lines.

Data quality report - Completeness omission ▶
Data quality measure reference
Measure description
The data reflects ongoing planned aerial survey tracts and is complete as of 7/12/24. This data will be updated as needed.

Lineage ▶

Lineage statement
asdsd

Process step ▶
When the process occurred 2025-04-30 00:00:00

Description

1. Data imported into GIS

Process contact - originator

Individual's name Deborah Brill
Organization's name RWSC
Contact's position Marine Mammals Subcommittee Lead

Contact information ▶

Address

Type
e-mail address deborah.brill@duke.edu

Process step ▶

When the process occurred 2025-04-30 00:00:00

Description

2. All features merged into single feature

Process contact - originator

Individual's name Deborah Brill
Organization's name RWSC
Contact's position Marine Mammals Subcommittee Lead

Contact information ▶

Address

Type
e-mail address deborah.brill@duke.edu

Process step ▶

When the process occurred 2025-04-30 00:00:00

Description

3. Field names added and filled in

Process contact - originator

Individual's name Deborah Brill
Organization's name RWSC
Contact's position Marine Mammals Subcommittee Lead

Contact information ▶

Address

Type
e-mail address deborah.brill@duke.edu

Source data ▶

Description

Received shapefiles directly from contacts

Distribution ▶

Distribution format

Name ⇔ Enterprise Geodatabase Feature Class

Transfer options

Transfer size ⇔ 0.000

Fields ▶

Details for object rpt.rpt.NEFSC_GOM_Aerial_Surveys ▶

Type ⇔ Feature Class
Row count ⇔ 6
Definition
Attribute table prepared by RWSC

Definition source
RWSC

Field objectid ▶

Alias ⇔ objectid
Data type ⇔ OID
Width ⇔ 4
Precision ⇔ 10
Scale ⇔ 0

Field description ⇔
Internal feature number.

Description source ⇔
Esri

Description of values ⇔
Sequential unique whole numbers that are automatically generated.

Field Shape ►

Alias ⇔ shape
Data type ⇔ Geometry
Width ⇔ 8
Precision ⇔ 0
Scale ⇔ 0

Field description ⇔
Feature geometry.

Description source ⇔
Esri

Description of values ⇔
Coordinates defining the features.

Field OPERATOR ►

Alias ⇔ OPERATOR
Data type ⇔ String
Width ⇔ 254
Precision ⇔ 0
Scale ⇔ 0

Field description
The primary affiliation for the operator of the device

Description source
RWSC

Description of values
Unique name affiliation

Field project_na ►

Alias ⇔ PROJECT_NA
Data type ⇔ String
Width ⇔ 254
Precision ⇔ 0
Scale ⇔ 0

Field POC_EMAIL ►

Alias ⇔ POC_EMAIL
Data type ⇔ String
Width ⇔ 254
Precision ⇔ 0
Scale ⇔ 0

Field description
Email for the primary point of contact

Description source
RWSC

Description of values
Unique email address

Field PATH_NAME ►

Alias ⇔ PATH_NAME
Data type ⇔ String
Width ⇔ 254
Precision ⇔ 0
Scale ⇔ 0

Field description
The route ID

Description source
RWSC

Description of values
Unique path identification

Field START_YEAR ►

Alias ⇔ START_YEAR
Data type ⇔ Integer
Width ⇔ 4
Precision ⇔ 10
Scale ⇔ 0

Field description

The start year in the YYYY format for the start of usable data for that path (i.e. the surveys are conducted in structured survey form).

Description source

RWSC

Description of values

Year in YYYY format

Field END_YEAR ►

Alias ⇔ END_YEAR

Data type ⇔ Integer

Width ⇔ 4

Precision ⇔ 10

Scale ⇔ 0

Field description

The end year in the YYYY format for the end of usable data for that path (i.e. the surveys are conducted in structured survey form). (*Default set to 2050 if no available end year)

Description source

RWSC

Description of values

Year in YYYY format

Field START_DATE ►

Alias ⇔ START_DATE

Data type ⇔ String

Width ⇔ 254

Precision ⇔ 0

Scale ⇔ 0

Field description

The start date in the D-Month format for the start of the typical survey season. (*NOT an actual date field, keeping it as text so that it doesn't add a default year)

Description source

RWSC

Description of values

Text start date for season

Field END_DATE ►

Alias ⇔ END_DATE

Data type ⇔ String

Width ⇔ 254

Precision ⇔ 0

Scale ⇔ 0

Field description

The end date in the D-Month format for the end of the typical survey season. (*NOT an actual date field, keeping it as text so that it doesn't add a default year)

Description source

RWSC

Description of values

Text end date for season

Field project_li ►

Alias ⇔ PROJECT_LI

Data type ⇔ String

Width ⇔ 254

Precision ⇔ 0

Scale ⇔ 0

Field date_submi ►

Alias ⇔ DATE_SUBMI

Data type ⇔ Date

Width ⇔ 8

Precision ⇔ 0

Scale ⇔ 0

Field FREQUENCY ►

Alias ⇔ FREQUENCY

Data type ⇔ String

Width ⇔ 254
Precision ⇔ 0
Scale ⇔ 0

Field description
Proposed/approximate frequency of surveys throughout the survey season

Description source
RWSC

Description of values
Unique frequency measure

Field lab ▶
Alias ⇔ LAB
Data type ⇔ String
Width ⇔ 254
Precision ⇔ 0
Scale ⇔ 0

Field DATE_ADDED ▶
Alias ⇔ DATE_ADDED
Data type ⇔ Date
Width ⇔ 8
Precision ⇔ 0
Scale ⇔ 0

Field description
The date the entries were added to map

Description source
RWSC

Description of values
Date field

Field st_length(shape) ▶
Alias ⇔ st_length(shape)
Data type ⇔ Double
Width ⇔ 0
Precision ⇔ 0
Scale ⇔ 0

Field description
field added by ESRI

Description source
ESRI

Description of values
field added by ESRI

Metadata Details ▶

Metadata language ⇔ English (UNITED STATES)
Metadata character set ⇔ utf8 - 8 bit UCS Transfer Format

Scope of the data described by the metadata ⇔ dataset
Scope name ⇔ dataset

Last update ⇔ 2026-08-12

ArcGIS metadata properties
Metadata format ArcGIS 1.0
Standard or profile used to edit metadata FGDC

Created in ArcGIS for the item 2024-01-24 12:31:40
Last modified in ArcGIS for the item 2026-08-12 13:38:36

Automatic updates
Have been performed Yes
Last update 2026-08-12 13:38:36

Item location history
Item copied or moved 2024-01-24 12:31:40
From C:\Users\jozog\OneDrive - HDR, Inc\GIS\HDR_NARW_Aerial_Tracklines
To \\DZ7YXT3\C\$\Users\jozog\OneDrive - HDR, Inc\GIS\Aerial_for_Jackie\HDR_NARW_Aerial_Tracklines

Metadata Contacts ▶

Metadata contact - originator

Individual's name Deborah Brill
Organization's name RWSC
Contact's position Marine Mammals Subcommittee Lead

Contact information ▶

Phone

Voice NA

Address

Type postal

City NA

Administrative area NA

Postal code NA

e-mail address deborah.brill@duke.edu

Metadata Maintenance ▶

Maintenance

Update frequency as needed